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## Tiling simply connected regions with rectangles



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Igor Pak, Jed Yang

Department of Mathematics, UCLA, Los Angeles, CA 90095, USA

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### ABSTRACT

In 1995, Beauquier, Nivat, Rémila, and Robson showed that tiling of general regions with two bars is NP-complete, except for a few trivial special cases. In a different direction, in 2005, Rémila showed that for simply connected regions by two rectangles, the tileability can be solved in quadratic time (in the area). We prove that there is a finite set of at most  $10^6$  rectangles for which the tileability problem of simply connected regions is NP-complete, closing the gap between positive and negative results in the field. We also prove that counting such rectangular tilings is #P-complete, a first result of this kind.

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## 1. Introduction

The study of *finite tilings* is a classical subject of interest in both theoretical and recreational literature [11,13]. In the *tileability problem*, a finite set of tiles  $\mathbf{T}$  is fixed, and a region is an input. This problem is known to be polynomial in some cases, and NP-complete in others (see [29]). Over the years, the hardness results were successively simplified (in statement, not in proof), with both sets of tiles and the regions becoming more restrictive. This paper is a new step in this direction.

In [3], it was shown that tiling of general regions with two bars is NP-complete, except for the case of dominoes. In a different direction, Rémila [36] (building on the ideas in [17,42]), showed that for *simply connected regions* and two rectangles, the tileability can be solved in quadratic time (in the area). The following theorem closes the gap between these polynomial and NP-complete results.

**Theorem 1.1 (Main Theorem).** *There exists a finite set  $\mathbf{R}$  of at most  $10^6$  rectangular tiles, such that the tileability problem of simply connected regions with  $\mathbf{R}$  is NP-complete.*

E-mail addresses: [pak@math.ucla.edu](mailto:pak@math.ucla.edu) (I. Pak), [jedyang@math.ucla.edu](mailto:jedyang@math.ucla.edu) (J. Yang).

There is a set of Wang tiles  $\mathbf{T}$  such that tiling simply connected regions with  $\mathbf{T}$  is NP-complete. We reduce Wang tileability to tileability with rectangular tiles. The reduction is *parsimonious* and is used to prove that counting the number of tilings of simply connected regions is also hard.

**Theorem 1.2.** *There exists a finite set  $\mathbf{R}$  of at most  $10^6$  rectangular tiles, such that counting the number of tilings of simply connected regions with  $\mathbf{R}$  is #P-complete.*

Although #P-completeness is known for tilings of general regions with right tromino and square tetromino [26], nothing was known previously for tilings with rectangles. We refer to Section 7 for the history of the problem, references, and further remarks.

## 2. Definitions and basic results

### 2.1. Ordinary tiles

Consider the integer lattice  $\mathbb{Z}^2$  as a union of closed unit squares with pairwise disjoint interiors. A region is a finite union of such unit squares such that the interior is connected. An (ordinary) tile is a finite simply connected region.

A tileset  $\mathbf{T}$  is a collection of tiles. Given a region  $\Gamma$  and a tileset  $\mathbf{T}$ , a  $\mathbf{T}$ -tiling of  $\Gamma$  is a union of translated copies of tiles from  $\mathbf{T}$  with pairwise disjoint interiors covering  $\Gamma$ . If a region admits a  $\mathbf{T}$ -tiling then it is  $\mathbf{T}$ -tileable. We may simply say *tiling* and *tileable* when  $\mathbf{T}$  is understood. Consider the following decision problems regarding tileability:

#### SIMPLY CONNECTED TILEABILITY

**Instance:** Simply connected region  $\Gamma$ , finite tileset  $\mathbf{T}$ .

**Decide:** Whether  $\Gamma$  is  $\mathbf{T}$ -tileable?

#### SIMPLY CONNECTED $\mathbf{T}$ -TILEABILITY

**Instance:** Simply connected region  $\Gamma$ .

**Decide:** Whether  $\Gamma$  is  $\mathbf{T}$ -tileable?

An input region can be given by the (finite) union of the squares it contains. The following is one of the early NP-completeness results [10].

**Theorem 2.1.** *If both region  $\Gamma$  and tileset  $\mathbf{T}$  are part of the input, SIMPLY CONNECTED TILEABILITY is NP-complete in the plane.*

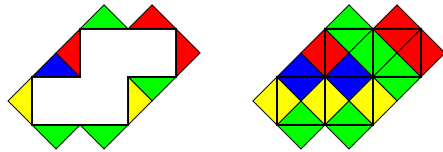
The following result is an extension of Theorem 2.1 proved by an explicit construction in an earlier version of this paper [30].

**Theorem 2.2.** *There exists a set  $\mathbf{T}$  of 23 tiles, such that SIMPLY CONNECTED  $\mathbf{T}$ -TILEABILITY is NP-complete.*

The history behind this theorem and its generalizations is outlined in Section 7.1.

### 2.2. Wang tiles

The edges of an ordinary tile are the unit-length edges on the boundary. Given a set of colors and an ordinary tile  $\tau$ , a *generalized Wang tile* is an assignment of colors to the edges of  $\tau$ . Note that an (ordinary) Wang tile is a generalized Wang tile of a unit square. The region  $\Gamma$  we are trying to tile will also have specified colors on its boundary. A region is (Wang) *tileable* if there is a tiling where incident edges have the same color, including on the boundary of the region (see Fig. 1). If a tileset consists of (generalized) Wang tiles, tileability always means Wang tileability.



**Fig. 1.** A colored region (left) and a Wang tiling (right). Colored edges are drawn as triangles for visibility.

### 2.3. Relational Wang tiles

Let us consider a more general setting. A set of *relational Wang tiles* is a collection  $\mathbf{W}$  of squares and the following data. The vertical (respectively horizontal) *Wang relation*  $V_{\mathbf{W}}(\tau, \tau')$  (respectively  $H_{\mathbf{W}}(\tau, \tau')$ ) specify that  $\tau' \in \mathbf{W}$  is allowed to be placed immediately below (respectively to the right of)  $\tau \in \mathbf{W}$ . We suppress the subscripts when it can be understood from context. The *boundary tiles* of a region  $\Gamma$  is a map from the exterior edges of  $\Gamma$  to the tiles  $\mathbf{W}$ . By abuse of language, we define the notion of tiling in this context: a  $\mathbf{W}$ -tiling of a region  $\Gamma$  is a map  $\pi : \Gamma \rightarrow \mathbf{W}$  such that tiles placed next to each other satisfy the Wang relations. Whenever a tile is adjacent to an exterior edge, we check the Wang relations as if the boundary tile corresponding to the edge is on the other side of the edge.

### 2.4. Decision and counting problems

Throughout the paper we consider many tiling problems that are NP-complete. All these problems are trivially in NP. Indeed, given a description of a tiling, one could simply check if it is in fact a tiling. To prove NP-hardness, we reduce a known NP-complete problem to the problem in question. We refer to [10,32] for definitions and details.

We also consider natural counting problems corresponding to the decision problems. For example, instead of asking whether satisfying assignments exist, we ask *how many* satisfying assignments there are. Similarly, for tileability, we count the number of tilings. If in the proof of NP-completeness, the corresponding reductions give a bijection between the sets of solutions, we call such reduction *parsimonious*.

Parsimonious reductions have the additional benefit of proving counting results using the same reduction. The class #P consists of the counting problems associated with decisions problems in NP. A counting problem is #P-complete if it is in #P and every #P question can be reduced to it. Thus, if there is a parsimonious reduction from problem  $Q_1$  to  $Q_2$ , then if  $Q_1$  is #P-complete, then so is  $Q_2$ . We refer to [43] (see also [32]) for definitions and details on #P complexity class.

## 3. Reduction lemmas

### 3.1. Basic reductions

In this section we consider five classes of TILEABILITY problems. Let  $\mathcal{T}$  be a collection of tiles and  $\mathcal{R}$  be a collection of regions. A decision problem in  $(\mathcal{T}, \mathcal{R})$ -TILEABILITY consists of a *fixed* tileset  $\mathbf{T} \subset \mathcal{T}$ , receives some  $\Gamma \in \mathcal{R}$  as input, and outputs whether  $\Gamma$  is  $\mathbf{T}$ -tileable.

We say  $(\mathcal{T}, \mathcal{R})$ -TILEABILITY is *linear time reducible* to  $(\mathcal{T}', \mathcal{R}')$ -TILEABILITY if for any finite tileset  $\mathbf{T} \subset \mathcal{T}$ , there exists a finite tileset  $\mathbf{T}' \subset \mathcal{T}'$  and a *reduction map*  $f : \mathcal{R} \rightarrow \mathcal{R}'$  that is computable in linear time (in the complexity of  $\Gamma \in \mathcal{R}$ ), such that  $\Gamma \in \mathcal{R}$  is  $\mathbf{T}$ -tileable if and only if  $f(\Gamma)$  is  $\mathbf{T}'$ -tileable.<sup>1</sup> If, moreover, that  $(\mathcal{T}', \mathcal{R}')$ -TILEABILITY is linear time reducible to  $(\mathcal{T}, \mathcal{R})$ -TILEABILITY, then they are *linear time equivalent*. Note that the transformation of the tilesets need not be efficient nor bijective.

For instance, if  $\mathcal{T}$  is the collection of all rectangular tiles and  $\mathcal{R}$  consists of simply connected regions, then  $(\mathcal{T}, \mathcal{R})$ -TILEABILITY is a class of problems regarding tiling simply connected regions with

<sup>1</sup> Recall that the tiles in the input are given as collections of unit squares.

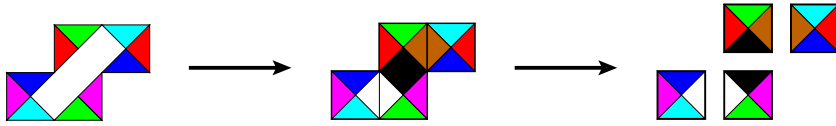


Fig. 2. From generalized Wang tiles to ordinary ones.

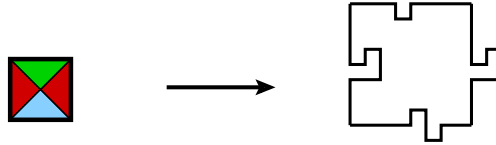


Fig. 3. Replacing each colored edge by a zig-zag curve to get ordinary tiles.

rectangular tiles. To simplify the notation, we drop the prefix in  $(\mathcal{T}, \mathcal{R})$ -TILEABILITY when the sets  $\mathcal{T}$  and  $\mathcal{R}$  are understood.

**Lemma 3.1** (Tileability Equivalence Lemma). *The following five classes of SIMPLY CONNECTED TILEABILITY problems are linear time equivalent:*

- (i) TILEABILITY with a fixed set of rectangular tiles.
- (ii) TILEABILITY with a fixed set of ordinary tiles.
- (iii) TILEABILITY with a fixed set of generalized Wang tiles.
- (iv) TILEABILITY with a fixed set of ordinary Wang tiles.
- (v) TILEABILITY with a fixed set of relational Wang tiles.

Moreover, the size of the tileset can be preserved in the reductions between (ii) and (iii).

**Proof.** The reductions  $(i) \Rightarrow (ii) \Leftrightarrow (iii) \Rightarrow (iv) \Rightarrow (v)$  are elementary and given below. The reduction  $(v) \Rightarrow (i)$  is stated separately as Lemma 3.4 and proved in the next section.

We may consider a rectangular tile as an ordinary tile, which in turn is a monochromatic generalized Wang tile. Therefore the reductions  $(i) \Rightarrow (ii) \Rightarrow (iii)$  are immediate, where each reduction map is simply the identity.

$(iii) \Rightarrow (iv)$ . Given a set of generalized Wang tiles, color each interior edge with a new color not used anywhere else, and consider each square as a separate ordinary Wang tile (see Fig. 2). These tiles are forced to reassemble themselves as the original generalized Wang tiles. The reduction map is again the identity.

$(iv) \Rightarrow (v)$ . It is obvious how to define the Wang relations to mimic the colored Wang tiles without increasing the number of tiles. To encode the boundary conditions, we may need to introduce less than  $4\chi$  tiles, where  $\chi$  is the number of colors permitted on the boundary. Indeed, to specify a color  $c$  on the top boundary, we need to choose an (arbitrary) tile whose bottom color is  $c$ . If no such tile exists, we must add a new tile to do so. If we do not involve the new tile in any Wang relations in the other directions, then it will never be used in the actual tiling, and thus will not affect tileability. We do the same for the other three directions.

The final reduction  $(v) \Rightarrow (i)$  is more difficult and is the content of Lemma 3.4 and proved in a later section.

To preserve the number of tiles in  $(iii) \Rightarrow (ii)$ , scale the generalized Wang tile and replace each colored edge by an appropriate rectilinear zig-zag curve to encode the matching rules (see Fig. 3 and [12]).  $\square$

### 3.2. Main reduction

**Lemma 3.2.** *There exists a set  $\mathbf{T}$  of generalized Wang tiles such that SIMPLY CONNECTED  $\mathbf{T}$ -TILEABILITY is NP-complete. Moreover, the associated counting problem is #P-complete.*

**Lemma 3.3.** *In the conditions of Lemma 3.2, one can take  $\mathbf{T}$  to have 23 generalized Wang tiles with total area 133 and using 9 colors.*

Proof of Lemma 3.2 is outlined in Section 6. Lemma 3.3 is proved by an explicit construction in [30] (see also [45]).

**Lemma 3.4 (Reduction Lemma).** *For a set  $\mathbf{W}$  of at most  $k$  (ordinary) Wang tiles with  $c$  (boundary) colors, there exists a set  $\mathbf{R}$  of at most  $8(k + 4c)^2$  rectangular tiles with the following property. Given a simply connected colored region  $\Gamma$ , there is a simply connected region  $\Gamma'$  such that  $\Gamma$  is  $\mathbf{W}$ -tileable if and only if  $\Gamma'$  is  $\mathbf{R}$ -tileable. Moreover, this reduction is parsimonious and can be computed in linear time.*

We may transform the set of 23 generalized Wang tiles afforded by the proof in [30] of Lemma 3.2, according to the procedure outlined in (iii)  $\Rightarrow$  (ii) of Lemma 3.1, in order to obtain Theorem 2.2 using 23 ordinary tiles. Similarly, using the transformation of Lemma 3.4, we conclude the result for rectangular tiles in Theorem 1.1 (see Section 5.1). Theorem 1.2 can be shown in the same way (see Section 5.2).

## 4. Proof of the Reduction Lemma (Lemma 3.4)

### 4.1. Basics

In this section, we provide a connection between Wang tiles and ordinary rectangular tiles (by making a reduction from the latter to the former). Recall that by Lemma 3.1, we can replace generalized Wang tiles with relational Wang tiles.

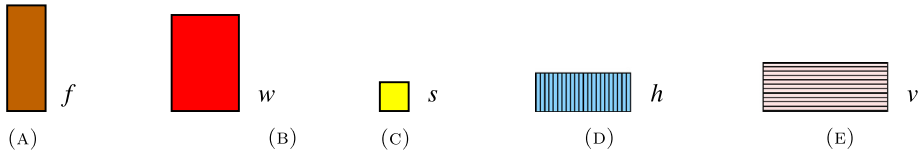
Without loss of generality, we may assume that the Wang relations are irreflexive, that is, there is no tile  $\tau$  such that  $H(\tau, \tau)$  or  $V(\tau, \tau)$ . Indeed, suppose  $\mathbf{W}$  is a set of Wang tiles. Let  $\mathbf{W}' = \{\tau_i: i \in \{0, 1\}, \tau \in \mathbf{W}\}$  be a doubled set of tiles. Define its horizontal Wang relation as follows. For  $\tau, \tau' \in \mathbf{W}$  and  $i, j \in \{0, 1\}$ , let  $H_{\mathbf{W}'}(\tau_i, \tau'_j)$  if and only if  $H_{\mathbf{W}}(\tau, \tau')$  and  $i \neq j$ . Its vertical Wang relation is defined analogously. It is clear that the Wang relations of  $\mathbf{W}'$  are irreflexive. Moreover, a  $\mathbf{W}$ -tiling can be made into a  $\mathbf{W}'$ -tiling by adding subscripts to the tiles in a checkerboard fashion, while the reverse can be done by ignoring the subscripts. Of course, the same transformation is done on the boundary tiles as well. Clearly this does not affect tileability nor the number of such tilings.

From now on, assume we are given a fixed set  $\mathbf{W}$  of relational Wang tiles whose relations  $H$  and  $V$  are irreflexive. Our goal is to produce a fixed set  $\mathbf{R}$  of rectangular tiles with the following property: Given any simply connected region  $\Gamma$  with specified boundary tiles, we can produce (in linear time) a simply connected region  $\Gamma'$  such that  $\Gamma$  is  $\mathbf{W}$ -tileable if and only if  $\Gamma'$  is  $\mathbf{R}$ -tileable. Moreover, the number of  $\mathbf{W}$ -tilings of  $\Gamma$  will be the same as the number of  $\mathbf{R}$ -tilings of  $\Gamma'$ .

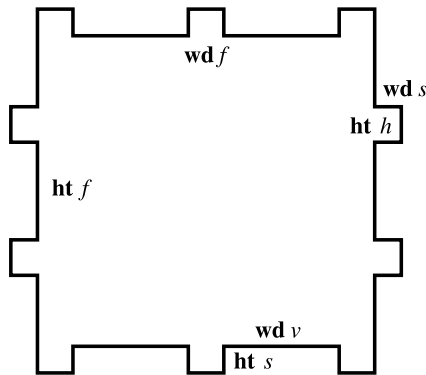
For simplicity, we first consider the case where we are given an  $r \times c$  rectangular region  $\Gamma$  with specified boundary tiles.

### 4.2. Expansion

From this point on, we only consider tiling using rectangular tiles. Fix  $M$  and  $e$  to be positive integers. Given a region  $\Gamma_0$ , we obtain an  $(M, e)$ -expansion  $\Gamma$  by scaling  $\Gamma_0$  by a factor of  $M$  and then *perturb* it by moving each corner vertex of the boundary curve of the region  $\Gamma$ , at most  $e$  in each direction, such that  $\Gamma$  is still a region (with rectilinear edges). Recall that a (rectangular) tile is just a simply connected region, thus the notion of  $(M, e)$ -expansion of a tile is defined. A tileset  $\mathbf{T}$  is an  $(M, e)$ -expansion of a tileset  $\mathbf{T}_0$  if each  $\tau \in \mathbf{T}$  is an  $(M, e)$ -expansion of some  $\tau_0 \in \mathbf{T}_0$ .



**Fig. 4.** Rectangular tiles  $\mathbf{R}_0$ : (A) fixed rectangle  $f$ , (B) fixed rectangle  $w$ , (C) flexible square  $s$ , (D) flexible rectangle  $h$ , and (E) flexible rectangle  $v$ .



**Fig. 5.** Boundary region  $\Gamma_0(2, 2)$ .

A tiling  $\pi$  of a region  $\Gamma$  is an  $(M, e)$ -expansion of a tiling  $\pi_0$  of some region  $\Gamma_0$  if it can be obtained by dilating by a factor of  $M$ , and then perturbing the tiles and the region by at most  $e$  as above. Note that after scaling, each tile may grow or shrink in each dimension by at most  $2e$ , and can shift around from its starting point by at most  $e$ .

Given a tiling  $\mathbf{T}_0$  and an  $(M, e)$ -expansion  $\mathbf{T}$ , a region  $\Gamma$  respects the expansion if there is a unique region  $\Gamma_0$  such that any  $\mathbf{T}$ -tiling of  $\Gamma$  is an  $(M, e)$ -expansion of a  $\mathbf{T}_0$ -tiling of  $\Gamma_0$ .

Intuitively, we will choose  $M > 100e$ , say, and carefully perturb only a few tiles, so that when consider tilings of regions respecting the expansion, we can essentially predict what the new tiling can be based on the original tiling.

#### 4.3. Rectangular tiles $\mathbf{R}_0$ and the region $\Gamma_0(r, c)$

Consider the following tiling:

$$\mathbf{R}_0 = \{f = R(34, 11), w = R(31, 14), s = R(10, 10), h = R(11, 31), v = R(14, 34)\},$$

where  $R(a, b)$  denotes a rectangle of height  $a$  and width  $b$  (see Fig. 4). For a rectangle  $t$ , write  $\mathbf{ht} t$  and  $\mathbf{wd} t$  for its height and width, respectively.

Now consider the region  $\Gamma_0(r, c)$  defined as follows (see Fig. 5). On each vertical side, there are  $r$  protrusions of height  $\mathbf{ht} h$  and width  $\mathbf{wd} s$ , separated by height  $\mathbf{ht} f$ . On each horizontal side, there are  $c$  cavities of width  $\mathbf{wd} v$  and height  $\mathbf{ht} s$ , separated by width  $\mathbf{wd} f$ .

**Sublemma 4.1.** The unique  $\mathbf{R}_0$ -tiling of  $\Gamma_0(r, c)$  consists of  $r$  rows and  $c$  columns of the  $w$  tile.

**Proof.** Fix natural numbers  $a = 10$  and  $b = 1$ . The tiles introduced above can now be written as  $f = R(3a+4b, a+b)$ ,  $w = R(3a+b, a+4b)$ ,  $s = R(a, a)$ ,  $h = R(a+b, 3a+b)$ , and  $v = R(a+4b, 3a+4b)$ .

We begin with a few definitions. A horizontal (vertical) segment of a region is called *bounded* if the region extends downward (to the right) on both sides of the segment. For  $t \in \{v, h\}$ , a pair  $(t, s)$  is the configuration of placing the tile  $s$  above or below  $t$ , aligned on the left. The *orientation* of the pair

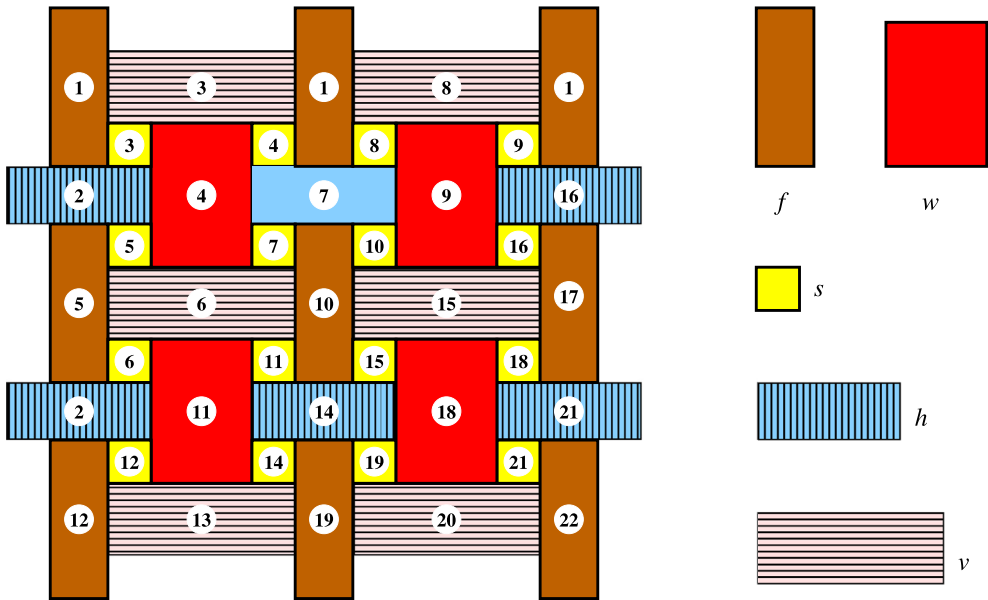


Fig. 6. Unique base tiling labeled by order.

is positive (negative) if  $s$  is placed below (above). Similarly, for  $t \in \{w, f\}$ , a pair  $(t, s)$  is obtained by placing  $s$  to the left or right of  $t$ , aligned on top. The orientation is positive (negative) if  $s$  is placed to the right (left). A bounded segment is *tilled by a tile (pair)* if in all tilings, the tile (pair) is adjacent to the segment.

We will tile the region  $\Gamma_0(r, c)$  in steps, as indicated by the numbers labeled on Fig. 6. Note that since  $a > b$ , each bounded horizontal segment of width  $\mathbf{wd} f$  on the top border must be tiled by  $f$  tiles, labeled 1. Similarly on the left, the bounded vertical segments of height  $\mathbf{ht} h$  must be tiled by  $h$  tiles, labeled 2. This creates a bounded vertical segment of height  $\mathbf{ht} v + \mathbf{ht} s$  on the top left corner; since  $a > 3b$ , it is tiled by the pair  $(v, s)$ , labeled 3. Since  $a > 4b$ , it is obvious that it needs to be positively oriented, to avoid a hole of width  $\mathbf{wd} v - \mathbf{wd} s$  and height  $\mathbf{ht} s$ , which cannot be filled.

Note that since  $a > 3b$ , this creates a new bounded horizontal segment of width  $\mathbf{wd} w + \mathbf{wd} s$ , which is tiled by the pair  $(w, s)$ , labeled 4. If  $w$  is on the left, it will create a bounded horizontal segment of width  $\mathbf{wd} f + \mathbf{wd} s$  to its left. Otherwise, if  $w$  is on the right, several  $s$  will be forced to appear on the left and still create the same bounded segment. Therefore, the  $(w, s)$  pair creates the bounded segment, regardless of how it is oriented.

Since  $a > 3b$ , this bounded horizontal segment of width  $\mathbf{wd} f + \mathbf{wd} s$  is again tiled by an  $(f, s)$  pair, labeled 5. Like the  $(v, s)$  pair above, since  $a > 4b$ , this needs to be positively oriented. This creates the bounded vertical segment of height  $\mathbf{ht} v + \mathbf{ht} s$ , tiled by a pair  $(v, s)$ , labeled 6, as above. In either orientation, it bounds the vertical segment of height  $\mathbf{ht} w$  above, concluding that the  $(w, s)$  pair (labeled 4) we placed above needs to be positively oriented. Furthermore, this bounds the vertical segment of height  $\mathbf{ht} h + \mathbf{ht} s$ , again tiled by the pair  $(h, s)$ , labeled 7. As before, in either orientation, we have a bounded vertical segment of height  $\mathbf{ht} v + \mathbf{ht} s$ , which necessarily needs to be tiled by the positively oriented pair  $(v, s)$ , labeled 8. This creates a bounded horizontal segment of width  $\mathbf{wd} w + \mathbf{wd} s$ .

We continue in like manner, working our way on the anti-diagonal from top right to bottom left. Each time we place the pair  $(w, s)$ , forcing the adjacent pair  $(h, s)$  placed in the previous stage to be positively oriented. Then we place  $(f, s)$ , forcing the adjacent  $(v, s)$  to be positively oriented as well. This procedure repeats with  $(w, s)$  and  $(f, s)$  in an alternating fashion. The last  $(f, s)$  will be placed in positive orientation and creates a bounded vertical segment of height  $\mathbf{ht} v + \mathbf{ht} s$ .

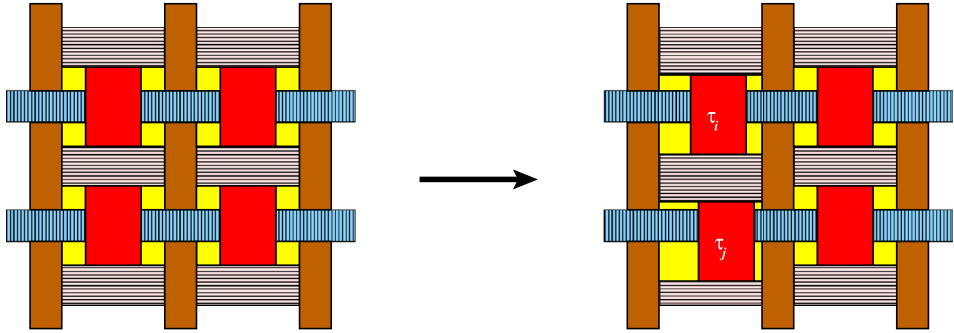


Fig. 7. Shifting an expansion of the unique tiling to represent Wang tiles.

Similarly, we work from bottom left to top right on the next anti-diagonal. We alternate between placing  $(v, s)$  and  $(h, s)$  pairs, positively orienting the  $(w, s)$  and  $(f, s)$  pairs in the previous stage, respectively. This continues until the entire region is filled.  $\square$

#### 4.4. Expansion $\mathbf{R}$ of $\mathbf{R}_0$

We will now define a clever set of perturbed expansion tiles that will correspond to the relational Wang tiles. Only the tiles  $s$ ,  $h$ , and  $v$  will have perturbations. Let  $\mathbf{W} = \{\tau_1, \dots, \tau_n\}$  be the fixed set of relational Wang tiles with irreflexive horizontal and vertical Wang relations  $H$  and  $V$ , respectively. Fix  $e = 5^n$  and  $M = 100e$  for the remainder of the section. Let  $\mathbf{R}$  be an  $(M, e)$ -expansion of  $\mathbf{R}_0$  as follows:

For  $t \in \{s, h, v\}$ , let  $t(a, b)$  be the scaled version of  $t$  with height and width increased by  $a$  and  $b$ , respectively. Imagine that the  $h$  and  $v$  tiles can stretch horizontally and vertically, respectively, and the  $s$  tiles can stretch in both directions. Then the  $w$  tiles, having no perturbations, will only shift around a little (by at most  $e$ ). The  $f$  tiles will stay fixed, enforcing the global structure. See Fig. 7. A  $w$  tile will be shifted to the right and down by  $5^i$  to represent the Wang tile  $\tau_i$ . To restrict the shifts to only those sizes, we replace  $s$  with the appropriate perturbed versions. Namely, for each  $i$ , introduce four tiles with perturbations  $s(\pm 5^i, \pm 5^i)$ , where all four combinations of signs are included. To enforce the Wang relations, for each  $\tau_i, \tau_j \in \mathbf{W}$  such that  $V(\tau_i, \tau_j)$  or  $H(\tau_i, \tau_j)$ , we introduce the perturbation  $v(5^j - 5^i, 0)$  or  $h(0, 5^j - 5^i)$ , respectively. This is the set  $\mathbf{R}$  we will use.

#### 4.5. Rectangular tiling

Obtain an  $(M, e)$ -expansion  $\Gamma(r, c)$  of  $\Gamma_0(r, c)$  by scaling with a factor of  $M$  and then perturbing it as follows. Recall that there are  $r$  protrusions on each vertical side and  $c$  cavities on each horizontal side. Each protrusion or cavity corresponds to a boundary tile of  $\Gamma$  in a natural way. Perturb the protrusion or cavity to the right or down, respectively, by  $5^i$  units if it corresponds to  $\tau_i$ .

**Sublemma 4.2.** *The  $(M, e)$ -expansion  $\Gamma(r, c)$  of  $\Gamma_0(r, c)$  respects the expansion  $\mathbf{R}$  of  $\mathbf{R}_0$ .*

**Proof.** Recall the argument in the proof of Sublemma 4.1. As the inequalities are all satisfied, the  $f$  tiles are fixed and force the perturbations to stay local. The  $w$  tiles have two degrees of freedom. They can move  $\pm 5^i$  in each direction, as regulated by the  $s$  tiles. Now note that the inequalities in the proof of Sublemma 4.1 are preserved. We leave the (easy) details to the reader.  $\square$

We now return to the proof of Lemma 3.4. It is clear that given a Wang  $\mathbf{W}$ -tiling of the rectangle  $\Gamma$  with boundary, we will get an  $\mathbf{R}$ -tiling of  $\Gamma(r, c)$ . Indeed, simply take the unique tiling of  $\Gamma_0(r, c)$  as afforded by Sublemma 4.1, scale by a factor of  $M$ , and then shift each  $w$  tile to the right and down by  $5^i$  if it represents  $\tau_i$ , and adjust the other tiles in the obvious way.



Conversely, if we are given an **R**-tiling of  $\Gamma(r, c)$ , we wish to recover the **W**-tiling of  $\Gamma$ . This is achieved using the following two sublemmas, both of which are clear when all numbers are considered in base 5; we omit the (easy) details.

**Sublemma 4.3.** *The equation  $5^i - 5^j = 5^k + 5^\ell$  does not admit a solution in  $\mathbb{N}$ .*

Therefore each  $w$  tile will shift to the right and down (as opposed to shifting left or up), and hence indeed represents a Wang tile  $\tau_i$  for some  $i$ .

**Sublemma 4.4.** *The equation  $5^i - 5^j = 5^k - 5^\ell$  does not admit solutions in  $\mathbb{N}$  except if  $i = j$  or  $i = k$ .*

If a  $w$  tile representing  $\tau_j$  is to the right of a  $w$  tile representing  $\tau_i$ , then  $h(0, 5^j - 5^i)$  must be in **R**. By the sublemma above, the differences  $5^j - 5^i$  are all distinct (recall that the Wang relations are irreflexive, so  $i = j$  does not happen), therefore we must have had  $H(\tau_i, \tau_j)$  as part of the Wang relation. The situation is similar for the vertical Wang relation  $V$ . So by reading off the associated tile  $\tau_i$  from the shifts of each  $w$  tile, we get a Wang **W**-tiling of  $\Gamma$ .

This completes the construction of  $\Gamma_0(r, c)$  for the case when  $\Gamma$  is a rectangle. For the general case, when  $\Gamma$  is a simply connected region, the proof follows verbatim after replacing  $\Gamma(r, c)$  and  $\Gamma_0(r, c)$  by appropriate regions.

It remains to get the upper bound estimates on the number of rectangles involved in the construction. Suppose we are given a set of  $k$  ordinary Wang tiles using  $c$  colors (on the boundary). By Lemma 3.1 we can equivalently consider a set of less than  $k + 4c$  relational Wang tiles. To satisfy irreflexivity, we might need to double the set of tiles, resulting in  $n = |\mathbf{W}| < 2(k + 4c)$  tiles. When making **R**, we will have one each of  $f$  and  $w$  tiles. There will be  $4n$  perturbed  $s$  tiles and at most  $n^2$  perturbed  $h$  and  $v$  tiles each. In total,

$$|\mathbf{R}| \leq 2n^2 + 4n + 2 = 2(n + 1)^2 \leq 8(k + 4c)^2.$$

This concludes the proof of Lemma 3.4.

## 5. Proof of theorems

### 5.1. Proof of Theorem 1.1

In the proof of Lemma 3.2 in [30], we constructed the set **W** of 23 generalized Wang tiles using 9 colors, such that SIMPLY CONNECTED **W**-TILEABILITY is NP-complete. It remains to count the total number of rectangles we obtain from the series of reduction constructions.

First, we compute the number of ordinary Wang tiles given by the transformation in Lemma 3.1. Observe that the total area of tiles in **W** is  $9 \cdot 5 + 8 \cdot 4 + 4 \cdot 14 = 133$ . Therefore we can break them into 133 ordinary Wang tiles by adding  $133 - 23$  more colors. But as these colors do not appear on the boundary, they need not be counted. Hence, in Lemma 3.4, we can take  $k = 133$  and  $c = 9$ , thus giving us at most  $10^6$  rectangles.

### 5.2. Proof of Theorem 1.2

Note that the reduction in the proof of Theorem 1.1 is parsimonious. Therefore as in the proof of Theorem 1.1, we apply Lemma 3.4 to conclude that  $10^6$  rectangles suffice.

## 6. Sketch of Lemma 3.2

Our proof of Lemma 3.2 in an earlier version [30] is completely elementary and yields explicit bounds (see also Section 7.1). However, the referee found it perhaps “too classical,” so even though no explicit bound was previously known, we agreed to omit it. On a positive note, a further reduced bound is given in [45] (see below).

Let us sketch an alternate proof of the lemma, using a non-deterministic *universal Turing machine* (UTM). It was suggested to us by Cris Moore.

Fix some non-deterministic universal Turing machine  $\mathfrak{M}$ . Given two finite tape configurations and a natural number  $t$  (in unary), it is NP-complete to decide whether  $\mathfrak{M}$  transforms the first tape configuration to the second with  $t$  steps of computation. Fix a finite set  $\mathbf{W}$  of Wang tiles that simulate the space–time computation diagram of  $\mathfrak{M}$  (see e.g. [22, §7]). Encode the given tape configurations as the top and bottom boundaries of a rectangular region with height  $t$ . This region is tileable by  $\mathbf{W}$  if and only if  $\mathfrak{M}$  transforms the first tape configuration to the second in precisely  $t$  steps. The details are straightforward.

Note that this method also proves the counting result. Indeed, one can devise a UTM so that there is a bijective correspondence between the accepting paths of the UTM and of the Turing machine it is simulating.

The proof in [30] constructs a set of 23 generalized Wang tiles (133 ordinary Wang tiles). However, it is possible to decrease these numbers by elementary means. After this paper was written, a modified construction by Günter Rote and the second author improves the number of generalized Wang tiles in Lemma 3.2 to 15, which amounts to 35 ordinary Wang tiles. With other technical improvements this does reduce the  $10^6$  bound in Theorem 1.1 to a much friendlier 117. The details are given in [45].

We do not know if this approach leads to improvements in the number of Wang tiles in the lemma, as this would depend on the smallest UTM. Given an  $m$ -state  $n$ -symbol Turing machine with  $k$  instructions, the standard construction of Wang tiles to simulate such a Turing machine yields more than  $nm + n + k$  tiles. By way of comparison, among the smallest known UTMs, this minimum is achieved by Rogozhin's 4-state 6-symbol machine with 22 instructions, which already yields more than 52 tiles [39] (see also [27]). Unless substantial progress is made in finding small UTMs, our elementary proof still gives better bounds.

## 7. Final remarks and open problems

7.1. Let us say a few brief words about the history of our results. Theorem 2.1 is classical. It was announced already in [10], referencing an unpublished preprint. Of course, now we have much stronger results.

A version of Lemma 3.2 was first announced in Levin's original 1973 short note regarding NP-completeness [21], but the proof has never been published.<sup>2</sup> Although we were unable to find in the literature an explicit construction for either Lemma 3.2 or, equivalently, of Theorem 2.2, we do not claim this result as ours, since it became a folklore decades ago. We include the proof for completeness, and since we need an explicit construction. An alternative proof is outlined in Section 6 above.

The general universal Turing machine approach goes back to Turing and von Neumann and is now classical. There are numerous applications to a range of combinatorial problems, although we have not seen the version discussed in Section 6.

Let us mention that using [28], the number of tiles in Theorem 2.2 can be reduced to 11, but this reduction has no effect on the number of tiles in the main theorems. Indeed, Theorem 2.2 is an immediate corollary of Lemma 3.2, which is the one needed in the proof of main Theorems 1.1 and 1.2.

7.2. In the tiling literature, the original theoretical emphasis was on tileability of the plane, the decidability and aperiodicity. The problem was often stated in the equivalent language of Wang tiles [4,38,44]. Unfortunately, there does not seem to be any standard treatment of the *finite Wang tiling* problems. Although some equivalences in Lemma 3.1 are routine, such as the reduction in Fig. 2, others seem to be new. We present full proofs for completeness.

<sup>2</sup> Leonid Levin, personal communication.

7.3. Historically, finite tilings were a backwater of the tiling theory, with coloring arguments being the only real tool [11]. On a negative (complexity) side, originally, the tileability problem was studied for general regions, where the tiles were part of the input. The NP-completeness of this most general problem is given in [10, §GP13]. When the set of tiles is fixed, NP-completeness was shown for general regions and various fixed small sets of tiles (see [26] and [3] building on the earlier unpublished work by Robson).

On the positive side, papers of Thurston [42] and Conway and Lagarias [8] introduced the *height function* and the *tiling group* interrelated approaches. The key underlying idea is the use of combinatorial group theory applied to the boundary word of the *simply connected regions*, so the tilings become *Van Kampen diagrams* of the corresponding tiling group. This approach allowed numerous applications to perfect matchings [6], tile invariants [18,25,33], tileability [40], various local move connectivity results [19,35], classical geometric problems [15], applications to colorings and mixing time [24], etc. More relevant to this paper, the breakthrough result by C. Kenyon and R. Kenyon [17] proved that tileability with bars of simply connected (s.c.) regions can be decided in polynomial time. This result was further extended to all pairs of rectangles by Rémila in [36], and by Korn [18] to an infinite family of *generalized dominoes*. Our Main Theorem puts an end to the hopes that these results can be extended to larger sets of rectangles.

Note also that having s.c. regions gives a speed-up for polynomial problems. For example, domino tileability is a special case of perfect matching, solvable in quadratic time on all planar bipartite graphs [23]. However, Thurston's algorithm is linear time (in the area), for all s.c. regions (see [6,42]).

7.4. We conjecture that in the Main Theorem (Theorem 1.1), the number of rectangles can be reduced down to 3, thus matching the lower bound (Rémila's tileability algorithm for the case of two rectangles). As a minor supporting evidence in favor of this conjecture, let us mention that the proofs in [17,36] are crucially based on *local move connectivity*, which fails for three general rectangles. In the absence of algebraic methods, there seems to be no other (positive) approach to tileability.

7.5. This result of Main Theorem can be contrasted with a large body of positive results on tiling rectangular regions with a fixed set of rectangles.

**Theorem 7.1.** (“Tiling rectangles with rectangles” Theorem [20].) *For every finite set  $\mathbf{R}$  of rectangular tiles, the tileability problem of an  $[M \times N]$  rectangle can be decided in  $O(\log MN)$  time.*

Note that Theorem 7.1 has linear time complexity for the rectangular regions written in *binary*. This result is based on the pioneer results by Barnes [1,2] applying commutative algebra, the *finite basis theorem* [9] (see also [34]), and the *transfer matrix method* (see e.g. [41, Ch. 4]).

It seems, tilings of rectangles have additional structure, which general regions do not have. See e.g. [5,7,37] for assorted results on the subject. On the other hand, when the tiles are part of the input, deciding tileability can be NP-hard, and the proof can be used to show that counting tilings is #P-hard. Note that the results in [20] only discuss tileability, not counting. It would be interesting to obtain general results on the local move connectivity and hardness of counting results for tilings of rectangular regions with rectangles.

7.6. Although counting perfect matchings in general graphs is #P-complete, for the grid graphs a *Pfaffian formula* gives a count for the number of domino tilings for any (not necessarily simply connected) region; this formula can be applied in polynomial time [23] (see also [16]). In a different direction, Moore and Robson [26] conjecture that already for two bars, the problem is #P-complete for general regions. They note that the corresponding reductions in [3,26] are not parsimonious. Thus, until now, the #P-completeness was open for any finite set of rectangular tiles, even for general regions.

We make a stronger conjecture that for every tileset  $\mathbf{T}$  of two bars  $[1 \times k]$  and  $[\ell \times 1]$ , where  $k, \ell \geq 2$ ,  $(k, \ell) \neq (2, 2)$ , the counting of tilings by  $\mathbf{T}$  of simply connected regions is #P-complete. In particular, the number  $10^6$  in Theorem 1.2 can be decreased to 2. There is no direct evidence in favor of this, except that the general combinatorial counting problems tend to be #P-complete unless there

is a special algebraic formula counting them. Furthermore, when it comes to tile counting, there seems to be no direct benefit of simple connectivity of the regions, so such result is likely to be equally hard as for general regions. We refer to [14] for the introduction and references.

7.7. Although Theorem 7.1 extends directly to bricks in higher dimensions [20], this is an exception rather than the rule. In fact, we recently showed that almost no other positive tileability results extend to higher dimensions, even Thurston's algorithm mentioned above (see [31]).

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